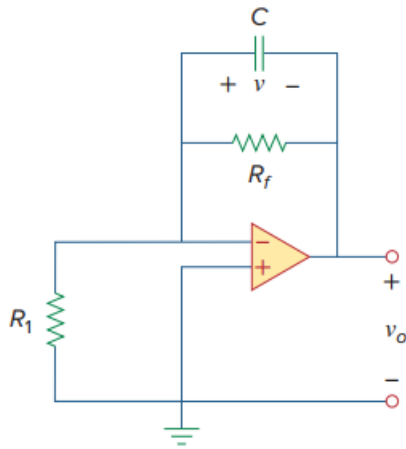


Problem

For the op amp circuit in Fig. 7.56, find v_o for $t > 0$ if $v(0) = 4$ V. Assume that $R_f = 50$ k Ω , $R_1 = 10$ k Ω , and $C = 10$ μ F.

Fig. 7.56:



Step-by-step solution

Step 1 of 4

Refer to Figure 7.56 in the textbook.

Voltage at non-inverting terminal is 0V.

Due to virtual short circuit between inverting and non-inverting terminals voltage at inverting terminal is 0V.

Apply Kirchhoff's voltage law at inverting terminal node.

$$\frac{0}{R_1} + \frac{v}{R_f} + C \frac{dv}{dt} = 0$$

$$C \frac{dv}{dt} = -\frac{v}{R_f}$$

$$\frac{dv}{v} = -\frac{dt}{R_f C} \dots\dots (1)$$

Step 2 of 4

Integrate the equation (1) on both sides.

$$\int \frac{dv}{v} = -\int \frac{dt}{R_f C}$$

$$\ln[v(t)] = -\frac{t}{R_f C} + \ln(A) \text{ (Here } A \text{ is a constant which result from above integration)}$$

$$\ln\left(\frac{v(t)}{A}\right) = -\frac{t}{R_f C}$$

$$v(t) = A e^{-\frac{t}{R_f C}} \dots\dots (2)$$

Step 3 of 4

Calculate the value of A using initial condition.

$$v(0) = 4\text{V}$$

Substitute $t = 0$ in equation (2)

$$v(0) = Ae^{-\frac{0}{R_f C}}$$

$$v(0) = A$$

$$A = 4$$

Therefore, the value of the voltage $v(t)$ is,

$$v(t) = 4e^{-\frac{t}{R_f C}} \text{ V} \quad t > 0 \text{ sec} \dots\dots (3)$$

Step 4 of 4

Substitute $50 \text{ k}\Omega$ for R , $10 \mu\text{F}$ for C in equation (3).

$$v(t) = 4e^{-\frac{t}{(50 \times 10^3)(10 \times 10^{-6})}}$$

$$= 4e^{-\frac{t}{0.5}} \text{ V} \quad \text{for } t > 0 \text{ sec}$$

From Figure 7.56, Observe that

$$0 - v = v_0$$

Substitute $4e^{-\frac{t}{0.5}} \text{ V}$ for v .

$$v_0 = -4e^{-\frac{t}{0.5}} \text{ V}$$

Therefore, the value of the v_0 is $-4e^{-2t} \text{ V}$ for $t > 0 \text{ sec}$.